Regression Analysis with example

Syllabus distribution

Unit	Content	Sub-content
4	Descriptive analysis	Mean, Median, mode, standard deviation, Co-variance, Coefficient of variation, correlation
5	Regression analysis and Inferential Analysis	Regression analysis, Linear Regression, Hypothesis testing, Tests
6	Model Making by Real life data	Model making using R

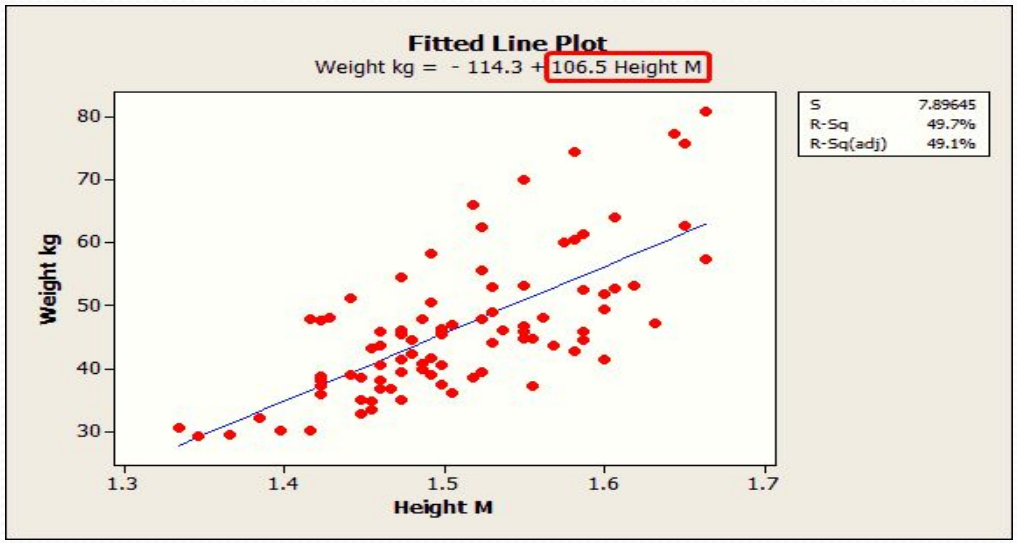
Unit-wise Marks Distribution

What is Regression Analysis

Regression is a supervised learning technique which helps in finding the correlation between variables and enables us to predict the continuous output variable based on the one or more predictor variables. It is mainly used for **prediction, forecasting, time series modeling, and determining the causal-effect relationship between variables**.

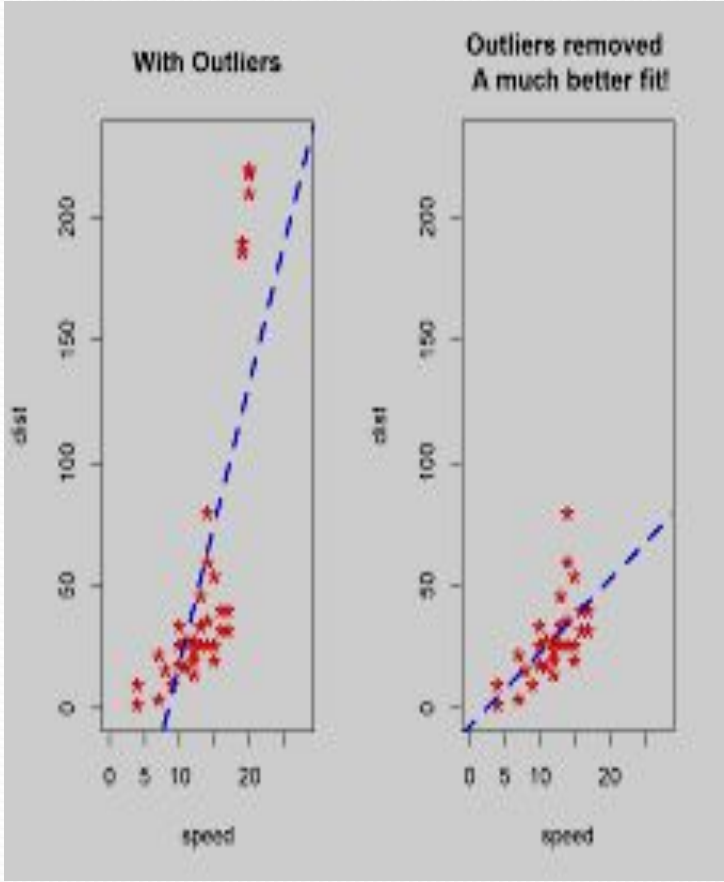
Examples of regression

- Prediction of rain using temperature and other factors
- Determining Market trends
- Prediction of road accidents due to rash driving.



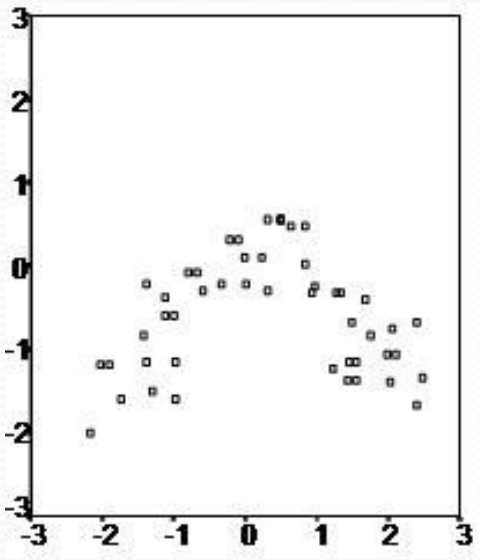
Terminologies Related to the Regression Analysis

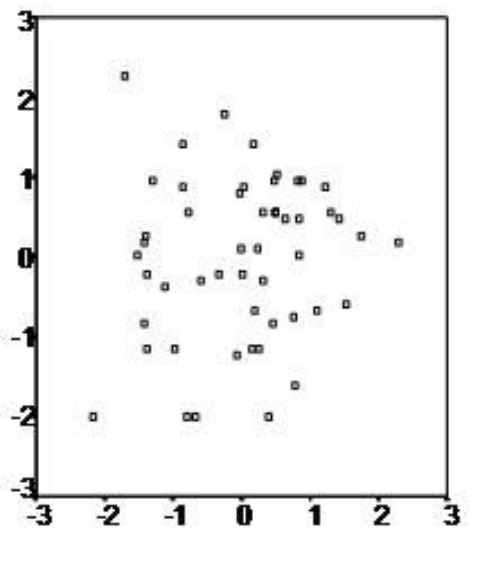
- **Dependent Variable:** The main factor in Regression analysis which we want to predict or understand is called the dependent variable. It is also called target variable.
- **Independent Variable**: The factors which affect the dependent variables or which are used to predict the values of the dependent variables are called independent variable, also called as a predictor.
- **Outliers**: Outlier is an observation which contains either very low value or very high value in comparison to other observed values. An outlier may hamper the result, so it should be avoided.



Assumptions for the Regression Analysis

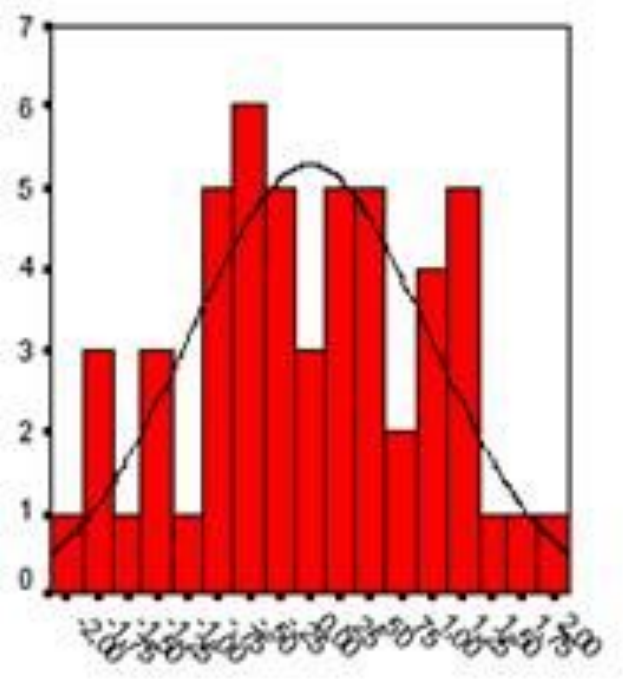
- Linear Relationship: The core premise of multiple linear regression is the existence of a linear relationship between the dependent (outcome) variable and the independent variables. This linearity can be visually inspected using scatterplots, which should reveal a straight-line relationship rather than a curvilinear one. (The linearity assumption can best be tested with scatter plots, the following two examples depict two cases, where no and little linearity is present.)





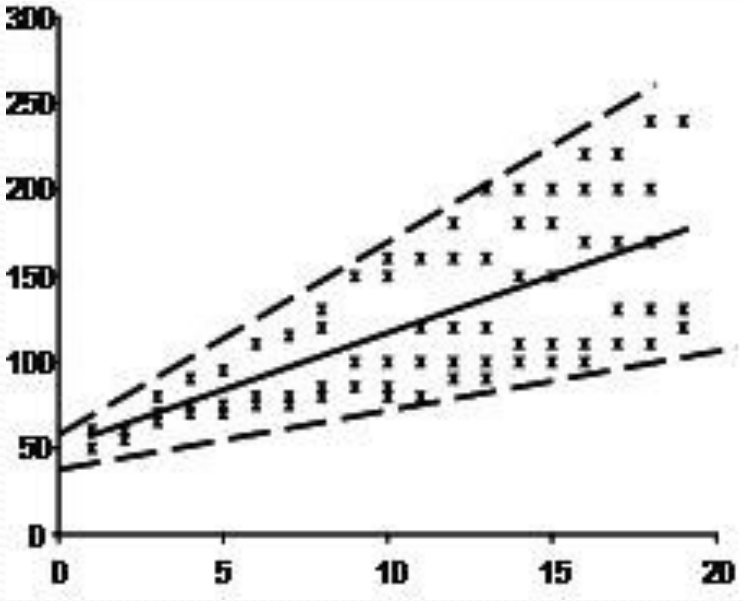
Assumptions for the Regression Analysis

- Multivariate Normality: The analysis assumes that the residuals (the differences between observed and predicted values) are normally distributed. This assumption can be assessed by examining histograms or Q-Q plots of the residuals, or through statistical tests such as the Kolmogorov-Smirnov test.



Terminologies Related to the Regression Analysis

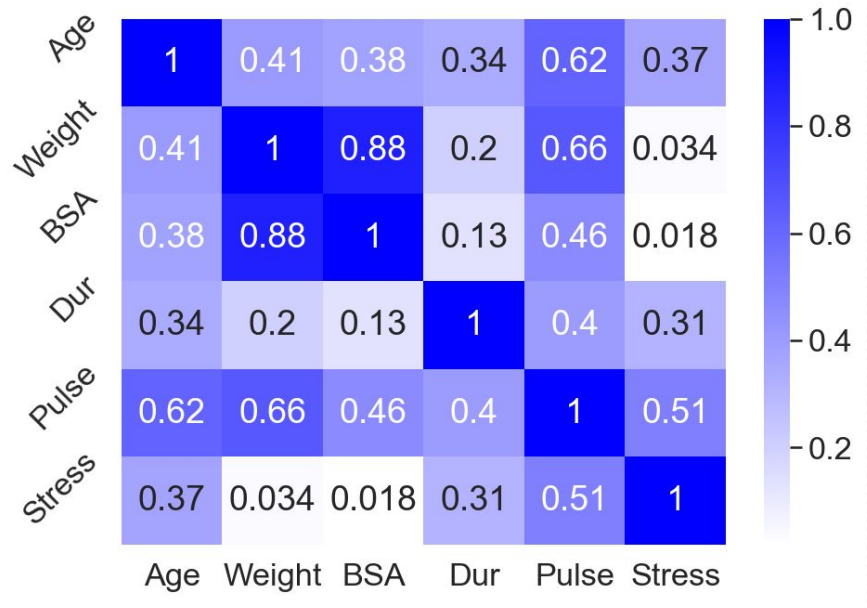
- Variance Inflation Factor (VIF), with VIF values above 10 indicating problematic multicollinearity. Solutions may include centering the data (subtracting the mean score from each observation) or removing the variables causing multicollinearity.



Assumptions for the Regression Analysis

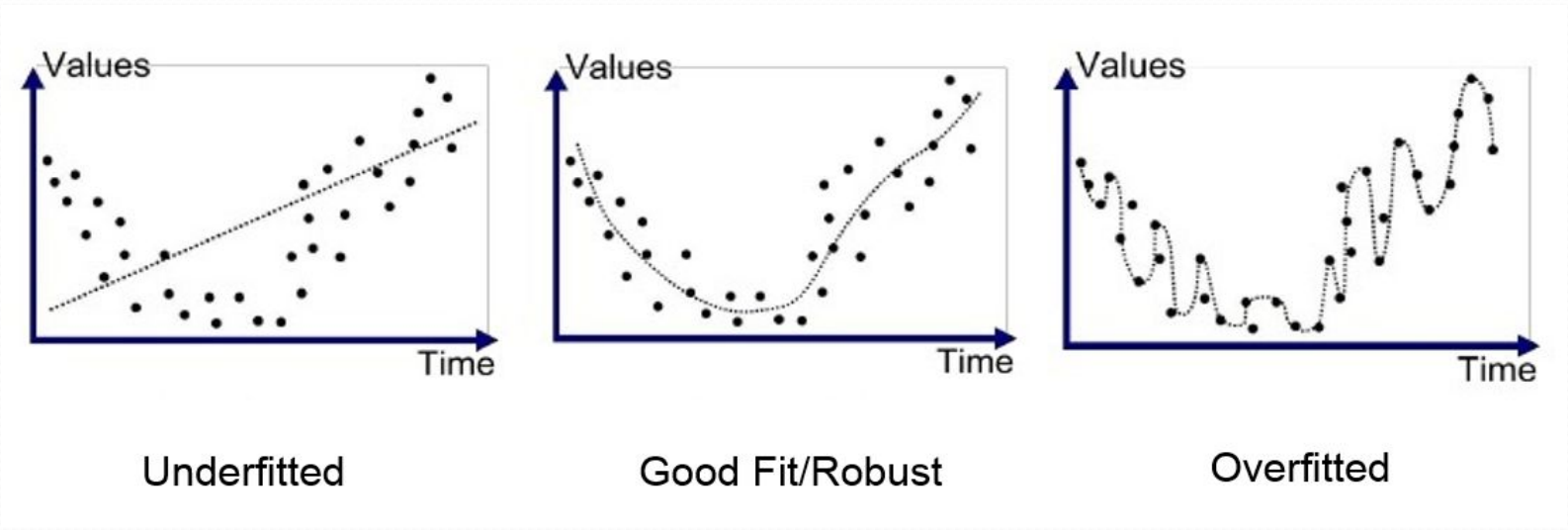
- **Multicollinearity**: If the independent variables are highly correlated with each other than other variables, then such condition is called Multicollinearity. It should not be present in the dataset, because it creates problem while ranking the most affecting variable.

(Weight & BSA are highly correlated)



Terminologies Related to the Regression Analysis

- **Underfitting and Overfitting**: If our algorithm works well with the training dataset but not well with test dataset, then such problem is called Overfitting. And if our algorithm does not perform well even with training dataset, then such problem is called underfitting.



Types of Regression

There are various types of regressions which are used in data science and machine learning. Each type has its own importance on different scenarios, but at the core, all the regression methods analyze the effect of the independent variable on dependent variables. Here we are discussing some important types of regression which are given below:

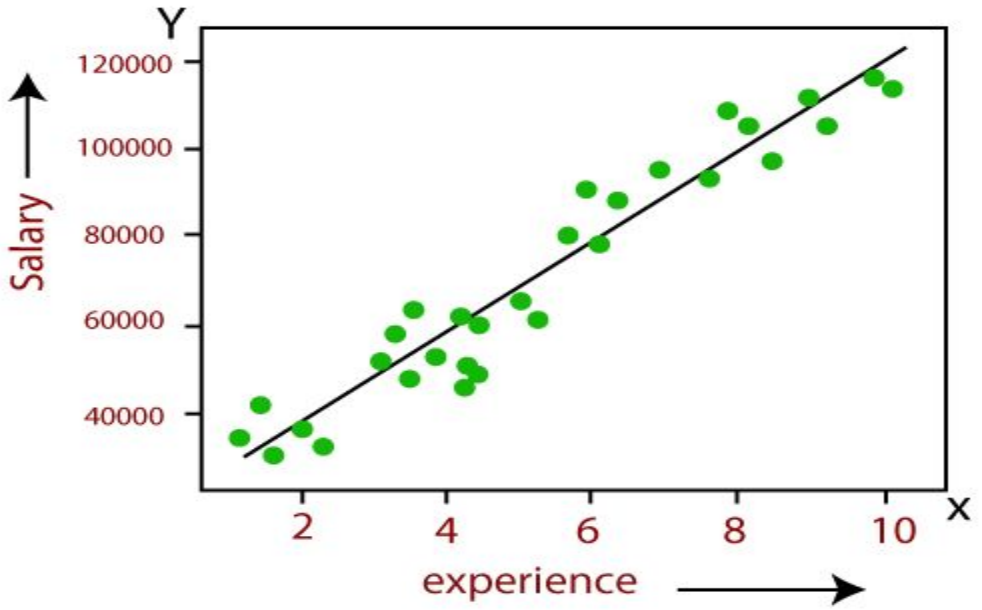
- **Linear Regression**
- **Logistic Regression**
- **Polynomial Regression**
- **Support Vector Regression**
- **Decision Tree Regression**
- **Random Forest Regression**
- **Ridge Regression**
- **Lasso Regression:**

Linear Regression

- Linear regression is a statistical regression method which is used for predictive analysis.
- It is one of the very simple and easy algorithms which works on regression and shows the relationship between the continuous variables.

Linear Regression

If there is only one input variable (x), then such linear regression is called **simple linear regression**. And if there is more than one input variable, then such linear regression is called **multipl**



Linear Regression

Equation of the Regression line is given by,

$Y = aX + b$

Here, Y = dependent variables (target variables),
X= Independent variables (predictor variables),
a and b are the linear coefficients

Some popular applications of linear regression are:

- Analyzing trends and sales estimates
- Salary forecasting
- Real estate prediction
- Arriving at ETAs in traffic.

Other Types of Regression

- **Logistic Regression:**

  Logistic regression is another supervised learning algorithm which is used to solve the classification problems. In **classification problems**, we have dependent variables in a binary or discrete format such as 0 or 1.

- **Polynomial Regression:**

  Polynomial Regression is a type of regression which models the non-linear dataset using a linear model.

  It is similar to multiple linear regression, but it fits a non-linear curve between the value of x and corresponding conditional values of y.

Other Types of Regression

- **Support Vector Regression:**

  Support Vector Machine is a supervised learning algorithm which can be used for regression as well as classification problems. So if we use it for regression problems, then it is termed as Support Vector Regression.

- **Decision Tree Regression:**

  Decision Tree is a supervised learning algorithm which can be used for solving both classification and regression problems.

Regression model for the data of Heart disease

Dataset Used

Dataset contains observations on the percentage of people biking to work each day, the percentage of people smoking, and the percentage of people with heart disease in an imaginary sample of 500 towns.

Regression model for the data of Heart disease

Step 1: Make sure your data meet the assumptions

- **Independence of observations**

Use the cor() function to test the relationship between your independent variables and make sure they aren't too highly correlated.

- **Normality**

Use the hist() function to test whether your dependent variable follows a normal distribution.

- **Linearity**

We can check this using two scatterplots: one for biking and heart disease, and one for smoking and heart disease

Regression model for the data of Heart disease

Step 2: Perform the linear regression analysis

```
> summary(Heart_disease.Model)

Call:
lm(formula = heart.disease ~ biking + smoking, data = Heart_Data)

Residuals:
    Min      1Q  Median      3Q     Max
-2.1789 -0.4463  0.0362  0.4422  1.9331

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 14.984658   0.080137  186.99   <2e-16 ***
biking      -0.200133   0.001366 -146.53   <2e-16 ***
smoking      0.178334   0.003539   50.39   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.654 on 495 degrees of freedom
Multiple R-squared:  0.9796,    Adjusted R-squared:  0.9795
```

```
> summary(Heart_disease.Model)

Call:
lm(formula = heart.disease ~ biking + smoking, data = Heart_Data)

Residuals:
    Min      1Q  Median      3Q     Max
-2.1789 -0.4463  0.0362  0.4422  1.9331

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 14.984658   0.080137  186.99   <2e-16 ***
biking      -0.200133   0.001366 -146.53   <2e-16 ***
smoking      0.178334   0.003539   50.39   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.654 on 495 degrees of freedom
Multiple R-squared:  0.9796,    Adjusted R-squared:  0.9795
```

- The estimated effect of biking on heart disease is -0.2, while the estimated effect of smoking is 0.178.
- This means that for every 1% increase in biking to work, there is a correlated 0.2% decrease in the incidence of heart disease. Meanwhile, for every 1% increase in smoking, there is a 0.178% increase in the rate of heart disease.

- **Multiple regression**: biking, smoking, and heart disease
- Let's see if there's a linear relationship between biking to work, smoking, and heart disease in our imaginary survey of 500 towns. The rates of biking to work range between 1 and 75%, rates of smoking between 0.5 and 30%, and rates of heart disease between 0.5% and 20.5%.
- To test the relationship, we first fit a linear model with heart disease as the dependent variable and biking and smoking as the independent variables. Run these two lines of code:

Step 5: Visualize the results with a graph

1. Create a new dataframe with the information needed to plot the model

- Create a sequence from the lowest to the highest value of your observed biking data;
- Choose the minimum, mean, and maximum values of smoking, in order to make 3 levels of smoking over which to predict rates of heart disease.

2. Predict the values of heart disease based on your linear model

3. Round the smoking numbers to two decimals

4. Change the 'smoking' variable into a factor

5. Plot the original data

6. Add the regression lines

7. Make the graph ready for publication

Final Conclusion

- Reporting the results of simple linear regressionWe found a significant relationship between income and happiness ($p < 0.001$, $R2 = 0.73 \pm 0.0193$), with a 0.73-unit increase in reported happiness for every $10,000 increase in income.
- Reporting the results of multiple linear regressionIn our survey of 500 towns, we found significant relationships between the frequency of biking to work and the frequency of heart disease and the frequency of smoking and frequency of heart disease ($p < 0$ and $p<0.001$, respectively).

- Specifically we found a 0.2% decrease ($\pm$ 0.0014) in the frequency of heart disease for every 1% increase in biking, and a 0.178% increase ($\pm$ 0.0035) in the frequency of heart disease for every 1% increase in smoking.